```
num_queries = 10   ❶

for i in range(num_queries):
    print("iteration", i)
    print("incumbent", train_x[train_y.argmax()], train_y.max())

    model, likelihood = fit_gp_model(train_x, train_y)   ❷

    policy = ...   ❸

    next_x, acq_val = botorch.optim.optimize_acqf(   ❹
        policy,   ❹
        bounds=torch.tensor([[-bound * 1.0], [bound * 1.0]]),   ❹
        q=1,   ❹
        num_restarts=20,   ❹
        raw_samples=50,   ❹
    )   ❹

    visualize_gp_belief_and_policy(model, likelihood, policy,
        next_x=next_x)   ❺

    next_y = forrester_1d(next_x)   ❻

    train_x = torch.cat([train_x, next_x])   ❻
    train_y = torch.cat([train_y, next_y])   ❻
```

❶ Number of evaluations of the objective function that can be made.

❷ Update the model on the current data.

❸ Initialize the BayesOpt policy, to be discussed later.

❹ Find the point that gives the highest acquisition score.

❺ Visualize the current GP model and acquisition scores.

❻ Make an observation at the identified point and update the training data.

With that, we have implemented a skeleton of a BayesOpt loop. All that's left for us to do is to fill in the initialization of `policy` with an actual BayesOpt policy that we would like to use, and the notebook will be able to run Bayesian optimization on the Forrester function. Note that while calling `visualize_gp_belief_and_policy()` isn't necessary (that is, the BayesOpt loop above will still be able to run without that line of code), the function is useful in helping us observe the behavior and characteristics of BayesOpt policies as well as diagnose any potential problems, as we will see later in this chapter.

Among the characteristics of a BayesOpt policy, one of the most important is the balance between exploration and exploitation, a classical tradeoff in many artificial intelligence and machine learning problems. Here, the possibility of discovering a high-performance region that we currently don't know about (exploration) is traded off against the chance of zeroing in on a known good region (exploitation). We will talk about this tradeoff in more detail in the next subsection.

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